General Quantum Sensing Notes and Interferometric Quantum Sensing

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1 General Notes

This will contain some general things to keep in mind when reading the rest of these notes.

- By "size of probe", we mean the number of sites it has. A probe of size L has L sites that can be occupied. Thus, L is the dimensionality of the system.

2 The Math of Quantum Sensing

Estimation is the final step in the process of metrology. After measurements on the system are performed, we need a mapping function to convert the outcomes we obtained into an estimate for the unknown parameter. This mapping function is called the *estimator*.

2.1 Classical Domain

- **Point Estimation** is the process in statistics where one uses data from a sample to find a single specific value, or "point", to serve as an "best guess" of a certain parameter, whose actual value is unknown. This guess is called a *point estimate*.
- **Definition:** A *point estimator* is any function $W(X_1, ..., X_n)$ of a sample $\{X_i\}_{i=1,..,N}$.
  It's a vague definition, as any statistic can serve as an estimator for a sample. A simple example of a point estimator is the sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$
- The *bias* of a point estimator W of a parameter θ is defined as the difference between the expected value of W and θ, i.e $\text{Bias}_\theta W = E_\theta[W] - \theta$. An estimator whose bias is identically 0 for all θ is called an *unbiased estimator* and satisfies $E_\theta[W] = \theta$
- The most widely used estimator is the Maximum Likelihood Estimator (MLE). If $\mathbf{X}$ is an iid sample with pdf $f(x|\theta)$, then the likelihood function is defined as $L(\theta|\mathbf{x}) = \prod_{i=1}^{n} f(x_i|\theta)$. This function is a measure of how likely it is that the parameter is θ, given the set of outcomes $\mathbf{x}$. Let $\hat{\theta}(\mathbf{x})$ be the parameter value at which $L(\theta|\mathbf{x})$ attains its maximum value. Then, the MLE of parameter θ based on sample $\mathbf{X}$ is $\hat{\theta}(\mathbf{X})$.
- The MLE is popular because it is *asymptotically unbiased and efficient*. This means that in the limit of $n \to \infty$, the MLE is unbiased and saturates the Cramer-Rao Inequality (efficient), which will be discussed later.
- In the case of metrology, the sample $\{X_1, ..., X_n\}$ is the set of random variables corresponding to n measurements of the system. The set $\{x_1, ..., x_n\}$, on the other hand, is the set of outcomes of those n measurements, i.e. the values of those n random variables.
- The sample $\{X_1, ..., X_n\}$ will be represented as $\mathbf{X}$, so the estimator $W(X_1, ..., X_n)$ will be written as $W(\mathbf{X})$. Similarly, $\{x_1, ..., x_n\}$ will be written as $\mathbf{x}$.
- Say θ is the parameter that we want to estimate. We are calculating $W(\mathbf{X})$ in an effort to get as close as possible to the true value of θ. Say $\mathbf{X}$ is the set of heights of n people. θ can be the true average height, then $W(\mathbf{X})$ will be the measured average height.
- From this example, you can see that changes in the parameter θ would change the Probability Density Function (PDF) of the sample $f(\mathbf{x})$. Hence, we say that the PDF of the sample is $f(\mathbf{x}|\theta)$.

2.1.1 Cramer-Rao Bound

Theorem: Let $\{X_1, ..., X_n\}$ be a sample with PDF $f(\mathbf{x}|\theta)$, and let $W(\mathbf{X})$ be any estimator such that

$$\frac{d}{d\theta}E_\theta W(\mathbf{X}) = \int_{\mathcal{X}} \frac{\partial}{\partial\theta}\big[W(\mathbf{x})f(\mathbf{x}|\theta)\big]\,d\mathbf{x} \qquad \text{and} \qquad \text{Var}_\theta W(\mathbf{X}) < \infty$$

Then

$$\boxed{\text{Var}_\theta(W(\mathbf{X})) \geq \frac{\left(\frac{d}{d\theta}E_\theta[W(\mathbf{X})]\right)^2}{E_\theta\left[\left(\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)\right)^2\right]}}$$

This is the **Cramer-Rao Bound**.
Before proving, let's explain the terms here.

- $E_\theta[W(\mathbf{X})]$ is the expectation value of the estimator $W(\mathbf{X})$ over the n random variables. $\text{Var}_\theta(W(\mathbf{X}))$ is the variance of the same.
- The θ subscript refers to the fact that the expectation value and variance are defined with the PDF $f(\mathbf{x}|\theta)$, instead of $f(\mathbf{x})$.
- The integral in the RHS is integrating over the sample space, denoted by $\mathcal{X}$, i.e. all possible values the data could take.
- The expression $\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)$ is called the **Score Function**.

Proof:
From Cauchy-Schwarz Inequality, for any two random variables X and Y,

$$[\text{Cov}(X,Y)]^2 \leq (\text{Var } X)(\text{Var } Y) \Rightarrow \text{Var } X \geq \frac{[\text{Cov}(X,Y)]^2}{\text{Var } Y}$$

Now let X be the estimator $W(\mathbf{X})$ and Y be the quantity $\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)$
We know that

$$\frac{d}{d\theta}E_\theta[W(\mathbf{X})] = \int_{\mathcal{X}} \frac{\partial}{\partial\theta}\big[W(\mathbf{x})f(\mathbf{x}|\theta)\big]\,d\mathbf{x} = \int_{\mathcal{X}} W(\mathbf{x})\left[\frac{\partial}{\partial\theta}f(\mathbf{x}|\theta)\right]d\mathbf{x} \quad \text{, since } W(\mathbf{x}) \text{ is independent of } \theta$$

Multiplying RHS by $f(\mathbf{x}|\theta)/f(\mathbf{x}|\theta)$ inside the integral, we get

$$\frac{d}{d\theta}E_\theta[W(\mathbf{X})] = \int_{\mathcal{X}} W(\mathbf{x})\frac{\left(\frac{\partial}{\partial\theta}f(\mathbf{x}|\theta)\right)}{f(\mathbf{x}|\theta)}f(\mathbf{x}|\theta)\,d\mathbf{x}$$

From the definition of Expectation value, we can say

$$\frac{d}{d\theta}E_\theta[W(\mathbf{X})] = E_\theta\left[W(\mathbf{X})\frac{\frac{\partial}{\partial\theta}f(\mathbf{X}|\theta)}{f(\mathbf{X}|\theta)}\right] = E_\theta\left[W(\mathbf{X})\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)\right]$$

Now, covariance is defined as $\text{Cov}(X,Y) = E[XY] - E[X]E[Y]$, but if we write the above equation after putting $W(\mathbf{X}) = 1$

$$\frac{d}{d\theta}E_\theta[1] = E_\theta\left[\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)\right]$$

$$\therefore \quad E_\theta\left[\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)\right] = 0$$

Thus, we can say

$$\text{Cov}_\theta\left(W(\mathbf{X}), \frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)\right) = E_\theta\left[W(\mathbf{X})\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)\right] - 0 = \frac{d}{d\theta}E_\theta[W(\mathbf{X})]$$

Also since $E_\theta\left[\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)\right] = 0$ we have

$$\text{Var}_\theta\left(\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)\right) = E_\theta\left[\left(\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)\right)^2\right] - 0$$

Now substituting these into the Cauchy-Schwarz Inequality:

$$\mathrm{Var}_\theta(W(\mathbf{X})) \geq \frac{\left(\frac{d}{d\theta}E_\theta[W(\mathbf{X})]\right)^2}{E_\theta\left[\left(\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)\right)^2\right]}$$

thus proving the theorem.

Special Case: iid Samples
In metrology, the different measurement outcomes can generally be assumed to be independent and identically distributed (iid). This allows us to simplify the Cramer-Rao Inequality. If the sample $\{X_1, ..., X_n\}$ is iid with PDF $f(x|\theta)$, then the denominator of the inequality can be simplified as

$$E_\theta\left[\left(\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)\right)^2\right] = nE_\theta\left[\left(\frac{\partial}{\partial\theta}\log f(X|\theta)\right)^2\right]$$

Therefore, the simplified Cramer-Rao Inequality is

$$\boxed{\mathrm{Var}_\theta(W(\mathbf{X})) \geq \frac{\left(\frac{d}{d\theta}E_\theta[W(\mathbf{X})]\right)^2}{nE_\theta\left[\left(\frac{\partial}{\partial\theta}\log f(X|\theta)\right)^2\right]}}$$

2.1.2 Fisher Information

The denominator of the RHS, $E_\theta\left[\left(\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)\right)^2\right]$, is called the *information number* of the sample, also called its ***Fisher Information***. It is a measure of how much information about θ is contained in the sample $\mathbf{X}$. It is a property of the sample, and has nothing to do with the chosen estimator. Since we deal with the iid case, we will work with Fisher Information of an outcome, denoted by $I(\theta)$ and defined as

$$\boxed{I(\theta) = E_\theta\left[\left(\frac{\partial}{\partial\theta}\log f(X|\theta)\right)^2\right]}$$

And Fisher Information of the sample will be $nI(\theta)$.

In the Cramer-Rao inequality, the numerator term $(\frac{d}{d\theta}E_\theta[W(\mathbf{X})])^2$ represents the sensitivity of the expectation of the estimator to changes in the parameter θ. If the estimator is unbiased, then it is equal to $(\frac{d}{d\theta}\theta)^2 = 1$. Since we're mostly working with MLE's, we can assume that n is large enough for the estimator to be approximately unbiased.
Since the estimator is unbiased, its variance is centred around θ. So from now on, we will represent variance of the estimator $\mathrm{Var}_\theta(W(\mathbf{X}))$ as $\mathrm{Var}(\theta)$. Of course, the variance of the actual parameter value θ is 0, as it is a fixed number. This just represents the value of the θ we're estimating.

Thus, we can rewrite the Cramer-Rao Bound for the iid and unbiased case:

$$\boxed{\mathrm{Var}(\theta) \geq \frac{1}{nI(\theta)}}$$

We can interpret the Cramer-Rao Bound as a relation between uncertainty of the estimator and information contained in the sample. If $\log f(X|\theta)$ varies sharply with θ, then $I(\theta)$ will be large, which means $\mathrm{Var}_\theta(W(\mathbf{X}))$ will have an even smaller lower bound. So, the more information contained in the sample, the more precise the estimator will be.
However, the variance of the estimator is always bounded by this inequality, so it cannot be arbitrarily precise.

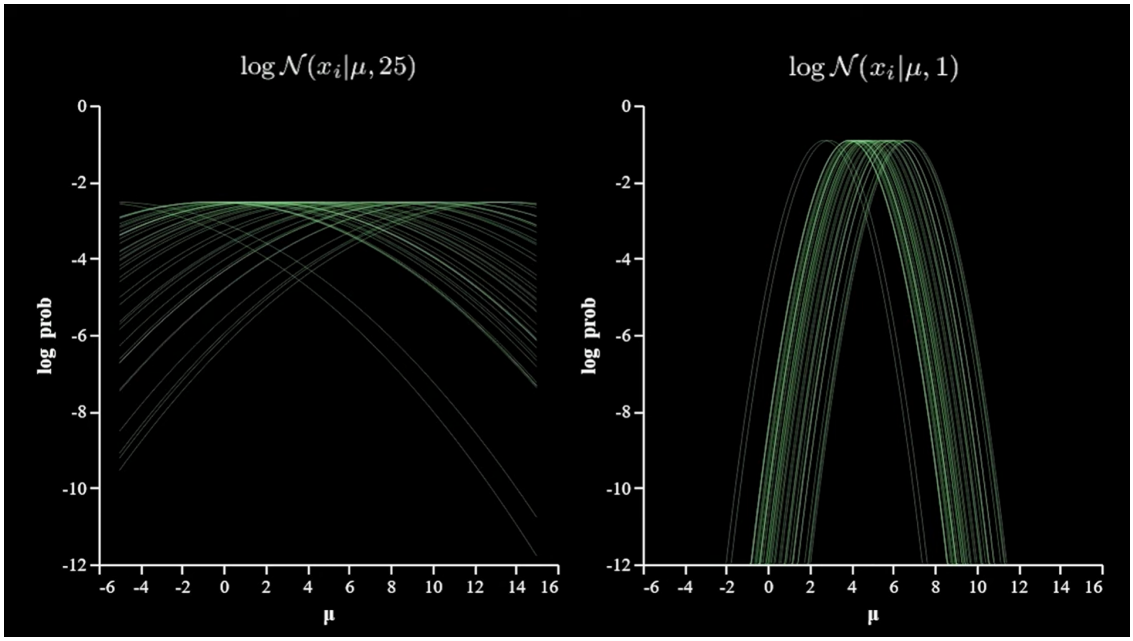


Figure 1

Above is a figure that gives us a visual way to show how larger Fisher Information allows more precise guesses of our parameter. These graphs are of the log-likelihood functions of a particular distributions, plotted against the parameter value, denoted here by μ. The left plot is for a Gaussian with large variance, and the right plot is for a Gaussian with small variance, each having the same fixed mean unknown to us. It is evident from this that it is much easier for us to estimate the value of μ from the plot on the right than the plot on the left. Thus we can see that when the slope is larger on average, each measurement gives us more information than when the slope is smaller on average.

2.1.3 A Interpretation of Fisher Information

Here, we'll see a more formal way of showing that the Fisher Information is a measure of how sharply the probability distribution changes with a change in θ.
Define a discrete probability distribution $P = \{p_i\}$, where each p_i is the probability of outcome x_i.

Shannon Information
First, we define the Shannon information of an outcome x_i for a probability distribution P as

$$h(x_i) = -\log p_i$$

Shannon Entropy
We define the Shannon Entropy $H(P)$ as the average Shannon Information of an outcome

$$H(P) = E[h(x_i)] = E[-\log p_i] = \sum_i -p_i \log p_i$$

In the continuous probability case, where the distribution has pdf $f(x)$, the Shannon entropy will be

$$H(f(x)) = \int_{\mathcal{X}} -f(x) \log f(x)\, dx$$

Kullback-Liebler Divergence
The Kullback-Lieber Divergence (KLD), represented by $D_{KL}(P||Q)$, is a measure of how different a probability distribution Q is compared to a reference distribution P (both defined for the same set of outcomes $\{x_i\}$). In the discrete case, it's defined as

$$D_{KL}(P||Q) = \sum_i p_i \log\left(\frac{p_i}{q_i}\right) = \sum_i p_i \log p_i - \sum_i p_i \log q_i$$

In the continuous case:

$$D_{KL}(f(x)||g(x)) = \int_{\mathcal{X}} f(x) \log f(x)\, dx \; - \int_{\mathcal{X}} f(x) \log g(x)\, dx$$

The first term is the negative Shannon Entropy of distribution P (or $f(x)$), the second term can be thought of as the "Cross Entropy" between distributions.
It is easy to see that $D_{KL}(P||Q) \neq D_{KL}(Q||P)$. The distribution chosen as the "reference" matters.

The Relation
Let's consider our estimation case, where we work with a sample with pdf $f(x|\theta)$. Let's find the KLD between $f(x|\theta)$ and $f(x|\theta + \Delta\theta)$.

$$D_{KL}(f(x|\theta)||f(x|\theta+\Delta\theta)) = \int_{\mathcal{X}} f(x|\theta)\log f(x|\theta)\,dx \; - \int_{\mathcal{X}} f(x|\theta)\log f(x|\theta+\Delta\theta)\,dx$$

Taking the Taylor expansion of the $\log f(x|\theta+\Delta\theta)$ term, we get

$$D_{KL} = \cancel{\int_{\mathcal{X}} f(x|\theta)\log f(x|\theta)\,dx} \; - \int_{\mathcal{X}} f(x|\theta)\left[\cancel{\log f(x|\theta)} + \Delta\theta\frac{\partial \log f(x|\theta)}{\partial\theta} + \frac{(\Delta\theta)^2}{2}\frac{\partial^2 \log f(x|\theta)}{\partial\theta^2} \; + \; ...\right]dx$$

$$\Rightarrow D_{KL} = -\Delta\theta\int_{\mathcal{X}} f(x|\theta)\frac{\partial \log f(x|\theta)}{\partial\theta}\,dx \; - \frac{(\Delta\theta)^2}{2}\int_{\mathcal{X}} f(x|\theta)\frac{\partial^2 \log f(x|\theta)}{\partial\theta^2}\,dx$$

Now, we can say that

$$\int_{\mathcal{X}} f(x|\theta)\frac{\partial \log f(x|\theta)}{\partial\theta}\,dx = E_\theta\left[\frac{\partial}{\partial\theta}\log f(X|\theta)\right] = 0$$

Taking derivative on both sides of that equation, we get

$$\frac{d}{d\theta}\int_{\mathcal{X}} f(x|\theta)\frac{\partial \log f(x|\theta)}{\partial\theta}\,dx = 0 \Rightarrow \int_{\mathcal{X}} \frac{\partial f(x|\theta)}{\partial\theta}\frac{\partial \log f(x|\theta)}{\partial\theta} \; + f(x|\theta)\frac{\partial^2 \log f(x|\theta)}{\partial\theta^2}\,dx = 0$$

$$\Rightarrow \int_{\mathcal{X}} \left(f(x|\theta)\frac{\partial \log f(x|\theta)}{\partial\theta}\right)\frac{\partial \log f(x|\theta)}{\partial\theta}\,dx \; + \int_{\mathcal{X}} f(x|\theta)\frac{\partial^2 \log f(x|\theta)}{\partial\theta^2}\,dx = 0$$

$$\Rightarrow \int_{\mathcal{X}} f(x|\theta)\frac{\partial^2 \log f(x|\theta)}{\partial\theta^2}\,dx = -\int_{\mathcal{X}} f(x|\theta)\left(\frac{\partial \log f(x|\theta)}{\partial\theta}\right)^2 dx = -E_\theta\left[\left(\frac{\partial}{\partial\theta}\log f(X|\theta)\right)^2\right] = -I(\theta)$$

Subsituting back into the D_{KL} equation, we finally get

$$\therefore \; D_{KL}(f(x|\theta)||f(x|\theta+\Delta\theta)) = \frac{1}{2}I(\theta)(\Delta\theta)^2$$

From this, we can see that the Fisher Information is a measure of how much the pdf changes with a small change in θ.

2.2 Quantum Domain

With the knowledge of classical Cramer-Rao Bound and Fisher Information, we now translate this knowledge into the Quantum realm.

- In Quantum Mechanics, according to the Born rule we have $f(x|\theta) = \mathrm{Tr}[\hat{E}_x\rho_\theta]$, where $\{\hat{E}_x\}$ are the elements of a Positive Operator-Valued Measure (POVM) and ρ_θ is the density operator, describing the state of the probe, parameterised by θ.
- The upper bound of the Fisher Information can be obtained by maximising the Fisher Information over all possible generalised measurements in quantum mechanics. This upper bound is the **Quantum Fisher Information (QFI)**

$$I_Q[\rho_\theta] = \max_{\{\hat{E}\}} I(\theta)$$

- This gives us the **Quantum Cramer-Rao (QCRB)**, which, for n independent measurements, is

$$\mathrm{Var}(\theta) \geq \frac{1}{nI_Q[\rho_\theta]}$$

- It is important to note that, since QFI is the maximised CFI over all measurements, it has no dependence on the actual measurement used in the process. It only depends on the state of the probe. This is in contrast to the CFI, which depends on the set of measurement outcomes.

For a general mathematical definition of QFI:

- Our first thought would be to get the quantum equivalent of the Score Function $\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)$. Since the equivalent of $f(X|\theta)$ is ρ_θ, one would think it would be $\frac{\partial}{\partial\theta}\log\rho_\theta = \rho_\theta^{-1}\frac{\partial\rho_\theta}{\partial\theta}$. This, however, becomes a problem when you realise that ρ_θ and its derivative do not generally commute (one can verify this by taking any positive Hermitian operator with unit trace). This would mean that the operator we defined is **not Hermitian**, which we cannot have (will not necessarily have real eigenvalues).

- We have the Left Logarithmic Derivative $\hat{L}_\mathrm{L} = \rho_\theta^{-1}\partial_\theta\rho_\theta \Rightarrow \rho_\theta\hat{L}_\mathrm{L} = \partial_\theta\rho_\theta$ and the Right Logarithmic Derivative $\hat{L}_\mathrm{R} = (\partial_\theta\rho_\theta)\rho_\theta^{-1} \Rightarrow \hat{L}_\mathrm{R}\rho_\theta = \partial_\theta\rho_\theta$. Neither of these are Hermitian.

- Thus, as a middle ground between the two, we define the **Symmetric Logarithmic Derivative (SLD) operator**, denoted by $\hat{L}$ which is implicitly defined by the equation

$$\frac{\partial\rho_\theta}{\partial\theta} = \frac{1}{2}(\rho_\theta\hat{L} + \hat{L}\rho_\theta)$$

  This is chosen to be the equivalent to $\frac{\partial}{\partial\theta}\log f(\mathbf{X}|\theta)$ in classical domain.

- We find that the expectation value of $\hat{L}$ is 0. We find this by taking trace on both sides of the above equation

$$\mathrm{Tr}\left[\frac{\partial\rho_\theta}{\partial\theta}\right] = \frac{1}{2}(2\ \mathrm{Tr}[\rho_\theta\hat{L}]) \Rightarrow \mathrm{Tr}[\rho_\theta\hat{L}] = \frac{\partial}{\partial\theta}\mathrm{Tr}[\rho_\theta] = \frac{\partial(1)}{\partial\theta} = 0$$

- Similar to the CFI, we get the QFI from the variance of the SLD operator.

$$I_Q[\rho_\theta] = \mathrm{Var}_\theta(\hat{L}) = \langle\hat{L}^2\rangle - \langle\hat{L}\rangle^2 = \mathrm{Tr}[\rho_\theta\hat{L}^2] - (\mathrm{Tr}[\rho_\theta\hat{L}])^2$$

$$\Rightarrow I_Q[\rho_\theta] = \mathrm{Tr}[\rho_\theta\hat{L}^2]$$

- If we know the eigenvalues λ_i and eigenstates $|i\rangle$ of ρ_θ, we can write its spectral decomposition $\rho_\theta = \sum_i \lambda_i |i\rangle\langle i|$. We can use this to get the matrix elements of $\hat{L}$ in the eigenbasis of ρ_θ. First we put $\langle m|$ and $|n\rangle$ on either side of the SLD definition

$$\langle m|\partial_\theta\rho_\theta|n\rangle = \frac{1}{2}\big(\langle m|\rho_\theta\hat{L}|n\rangle + \langle m|\hat{L}\rho_\theta|n\rangle\big)$$

$$\Rightarrow \langle m|\partial_\theta\rho_\theta|n\rangle = \frac{\lambda_m + \lambda_n}{2}\langle m|\hat{L}|n\rangle$$

$$\Rightarrow \hat{L}_{mn} = \frac{2}{\lambda_m + \lambda_n}\langle m|\partial_\theta\rho_\theta|n\rangle$$

- Using this, we can obtain a general expression for QFI

$$I_Q[\rho_\theta] = \mathrm{Tr}[\rho_\theta\hat{L}^2] = \sum_i \langle i|\rho_\theta\hat{L}^2|i\rangle = \sum_i \lambda_i\langle i|\hat{L}^2|i\rangle$$

  Using the completeness relation $\sum_j |j\rangle\langle j| = I$, we can do

$$I_Q[\rho_\theta] = \sum_{i,j}\lambda_i\langle i|\hat{L}|j\rangle\langle j|\hat{L}|i\rangle = \sum_{i,j}\lambda_i|\hat{L}_{ij}|^2$$

$$\Rightarrow I_Q[\rho_\theta] = \sum_{i,j}\frac{4\lambda_i}{(\lambda_i+\lambda_j)^2}|\langle i|\partial_\theta\rho_\theta|j\rangle|^2$$

  Since we are summing over all (i,j) pairs, we can swap i and j in this equation and get the same result. Thus, it will be more convenient to symmetrise it by taking the average of the 2 cases:

$$I_Q[\rho_\theta] = \frac{1}{2}\sum_{i,j}\frac{4(\lambda_i+\lambda_j)}{(\lambda_i+\lambda_j)^2}|\langle i|\partial_\theta\rho_\theta|j\rangle|^2 = \sum_{i,j}\frac{2}{(\lambda_i+\lambda_j)}|\langle i|\partial_\theta\rho_\theta|j\rangle|^2$$

  Now, since ρ_θ is positive semi-definite, $\lambda_i + \lambda_j \geq 0$. What will happen in the $\lambda_i + \lambda_j = 0 \Rightarrow \lambda_i = \lambda_j = 0$ case? Looking at the earlier expression $\sum_{i,j}\lambda_i|\hat{L}_{ij}|^2$, we see this will clearly vanish if $\lambda_i = 0$. Hence those terms must be excluded. Finally, we get a general expression for QFI, using the eigenvalues and eigenstates of the density matrix.

$$I_Q[\rho_\theta] = \sum_{\substack{i,j \\ \lambda_i+\lambda_j>0}}\frac{2}{\lambda_i+\lambda_j}|\langle i|\partial_\theta\rho_\theta|j\rangle|^2$$

2.2.1 Standard Quantum Limit (SQL)

We have already seen that for n independent measurements, the total QFI obtained will be n times the QFI of a single measurement. Thus, from the QCRB, $\text{Var}(\theta)$ scales proportionately with $1/n$. This is the **Standard Quantum Limit (SQL)**.
It is easy to see how this limit would also apply when the probe state ρ_θ is made up of N separable (not entangled) qubits. Since they are not entangled, measuring the state as a whole is like conducting N independent measurements. Hence, the SQL is followed in this case. If this measurement is conducted K times, the QCRB is $\text{Var}(\theta) \geq 1/KN$.

2.2.2 Heisenberg Limit

To explain the HL, we'll construct a system that achieves it.
We'll have to start with some assumptions.

- We assume the parameter-dependent transformation of the state is in the form of a Unitary operator $\hat{U}(\theta) = e^{-i\hat{G}\theta}$, where $\hat{G}$ represents the Hermitian *generator* of the parameter transformation. Examples of generators are the photon number operator, the angular momentum operators, etc.
- This unitary transforms the probe state like so: $\rho_\theta = e^{-i\hat{G}\theta}\rho_0 e^{i\hat{G}\theta}$, where ρ_0 is the initial state of the probe.
- We assume the probe state is pure, given by $\rho_\theta = |\psi_\theta\rangle\langle\psi_\theta|$. So the transformation is given by $|\psi_\theta\rangle = e^{-i\hat{G}\theta}|\psi_0\rangle$

We now re-evaluate the SLD equation

$$\frac{\partial\rho_\theta}{\partial\theta} = \frac{1}{2}(\rho_\theta\hat{L} + \hat{L}\rho_\theta)$$

Using the fact that

$$\frac{\partial\rho_\theta}{\partial\theta} = |\partial_\theta\psi_\theta\rangle\langle\psi_\theta| + |\psi_\theta\rangle\langle\partial_\theta\psi_\theta|$$

Subsituting this in, we get

$$|\partial_\theta\psi_\theta\rangle\langle\psi_\theta| + |\psi_\theta\rangle\langle\partial_\theta\psi_\theta| = \frac{1}{2}(|\psi_\theta\rangle\langle\psi_\theta|\hat{L} + \hat{L}|\psi_\theta\rangle\langle\psi_\theta|)$$

Multiplying this equation with $|\psi_\theta\rangle$ on the right and with $\langle\psi_\theta|$ on the left, we get the 2 equations

$$|\partial_\theta\psi_\theta\rangle\langle\psi_\theta|\psi_\theta\rangle + |\psi_\theta\rangle\langle\partial_\theta\psi_\theta|\psi_\theta\rangle = \frac{1}{2}(|\psi_\theta\rangle\langle\psi_\theta|\hat{L}|\psi_\theta\rangle + \hat{L}|\psi_\theta\rangle\langle\psi_\theta|\psi_\theta\rangle)$$

$$\langle\psi_\theta|\partial_\theta\psi_\theta\rangle\langle\psi_\theta| + \langle\psi_\theta|\psi_\theta\rangle\langle\partial_\theta\psi_\theta| = \frac{1}{2}(\langle\psi_\theta|\psi_\theta\rangle\langle\psi_\theta|\hat{L} + \langle\psi_\theta|\hat{L}|\psi_\theta\rangle\langle\psi_\theta|)$$

Using the normalised property of wavefunctions, i.e. $\langle\psi_\theta|\psi_\theta\rangle = 1$, and that expectation value of $\hat{L}$ is 0, we get

$$|\partial_\theta\psi_\theta\rangle + |\psi_\theta\rangle\langle\partial_\theta\psi_\theta|\psi_\theta\rangle = \frac{1}{2}\hat{L}|\psi_\theta\rangle \quad \text{and} \quad \langle\psi_\theta|\partial_\theta\psi_\theta\rangle\langle\psi_\theta| + \langle\partial_\theta\psi_\theta| = \frac{1}{2}\langle\psi_\theta|\hat{L}$$

$\hat{L}$ must satisfy both of the above equations. We obtain SLD operator as

$$\hat{L} = 2(|\partial_\theta\psi_\theta\rangle\langle\psi_\theta| + |\psi_\theta\rangle\langle\partial_\theta\psi_\theta|)$$

We can substitute it into the equation to verify.
Now, the QFI is the expectation value of $\hat{L}^2$, so

$$\hat{L}^2 = 4(|\partial_\theta\psi_\theta\rangle\langle\psi_\theta| + |\psi_\theta\rangle\langle\partial_\theta\psi_\theta|)(|\partial_\theta\psi_\theta\rangle\langle\psi_\theta| + |\psi_\theta\rangle\langle\partial_\theta\psi_\theta|)$$

$$\hat{L}^2 = 4(|\partial_\theta\psi_\theta\rangle\langle\psi_\theta|\partial_\theta\psi_\theta\rangle\langle\psi_\theta| + |\partial_\theta\psi_\theta\rangle\langle\partial_\theta\psi_\theta| + |\psi_\theta\rangle\langle\partial_\theta\psi_\theta|\partial_\theta\psi_\theta\rangle\langle\psi_\theta| + |\psi_\theta\rangle\langle\partial_\theta\psi_\theta|\psi_\theta\rangle\langle\partial_\theta\psi_\theta|)$$

$$\langle\hat{L}^2\rangle = \langle\psi_\theta|\hat{L}^2|\psi_\theta\rangle = 4(\langle\psi_\theta|\partial_\theta\psi_\theta\rangle\langle\psi_\theta|\partial_\theta\psi_\theta\rangle + |\langle\psi_\theta|\partial_\theta\psi_\theta\rangle|^2 + \langle\partial_\theta\psi_\theta|\partial_\theta\psi_\theta\rangle + \langle\partial_\theta\psi_\theta|\psi_\theta\rangle\langle\partial_\theta\psi_\theta|\psi_\theta\rangle)$$

We now need to use the fact that since $\partial_\theta\langle\psi_\theta|\psi_\theta\rangle = 0 \Rightarrow \langle\partial_\theta\psi_\theta|\psi_\theta\rangle = -\langle\psi_\theta|\partial_\theta\psi_\theta\rangle$.Using that, we get

$$\langle\hat{L}^2\rangle = 4(\cancel{-|\langle\psi_\theta|\partial_\theta\psi_\theta\rangle|^2} + \cancel{|\langle\psi_\theta|\partial_\theta\psi_\theta\rangle|^2} + \langle\partial_\theta\psi_\theta|\partial_\theta\psi_\theta\rangle - |\langle\psi_\theta|\partial_\theta\psi_\theta\rangle|^2)$$

Therefore, we get

$$I_Q = \langle\hat{L}^2\rangle = 4[\langle\partial_\theta\psi_\theta|\partial_\theta\psi_\theta\rangle - |\langle\psi_\theta|\partial_\theta\psi_\theta\rangle|^2]$$

Using the unitary transformation, we can say that $|\partial_\theta \psi_\theta\rangle = \partial_\theta |\psi_\theta\rangle = \partial_\theta e^{-i\hat{G}\theta}|\psi_0\rangle = -i\hat{G}e^{-i\hat{G}\theta}|\psi_0\rangle = -i\hat{G}|\psi_\theta\rangle$.
Similarly, $\langle\partial_\theta\psi_\theta| = i\langle\psi_\theta|\hat{G}$.
Substituting these in the above eqn, we get

$$I_Q = 4[i(-i)\langle\psi_\theta|\hat{G}^2|\psi_\theta\rangle - |(-i)\langle\psi_\theta|\hat{G}|\psi_\theta\rangle|^2]$$

$$I_Q = 4[\langle\hat{G}^2\rangle - \langle\hat{G}\rangle^2] = 4\ \mathrm{Var}(\hat{G})$$

So as the variance of $\hat{G}$ increases, the QFI increases, and thus the variance lower-bound of θ decreases.

NOTE:
In the general case, the expectation values of $\hat{G}$ and $\hat{G}^2$, are calculated with respect to the post-transformation state $|\psi_\theta\rangle$. However, in the case of unitary transformations, we can use the fact that $[\hat{G}, e^{-i\hat{G}\theta}] = 0$ (as linear operators commute with functions of themselves), to obtain

$$\langle\hat{G}\rangle_\theta = \langle\psi_\theta|\,\hat{G}\,|\psi_\theta\rangle = \langle\psi_0|\,e^{i\hat{G}\theta}\hat{G}e^{-i\hat{G}\theta}\,|\psi_0\rangle = \langle\psi_0|\,e^{i\hat{G}\theta}e^{-i\hat{G}\theta}\hat{G}\,|\psi_0\rangle = \langle\psi_0|\,\hat{G}\,|\psi_0\rangle = \langle\hat{G}\rangle_0$$

Similarly, $\langle\hat{G}^2\rangle_\theta = \langle\hat{G}^2\rangle_0$.
So the variances are the same.
END OF NOTE

Let the probe state be multipartite, made up of N particles. Then the N-qubit unitary acting on the system will be of the form $\hat{U}(\theta) = e^{-i\hat{g}\theta} \otimes e^{-i\hat{g}\theta} \otimes ...$, where $\hat{g}$ is the generator on each qubit. The total generator $\hat{G}$, then, will be the sum of individual generators $\hat{G} = \sum_{i=1}^{N} \hat{g}_i$.
Here, $\hat{g}_i$ is an N-qubit gate defined as $\hat{g}_i = \mathbb{I} \otimes \mathbb{I} \otimes ... \otimes \hat{g} \otimes ... \otimes \mathbb{I}$, where $\hat{g}$ operates on the ith qubit and $\mathbb{I}$ operates on the other qubits.
Let the maximum and minimum eigenvalues of the $\hat{g}$ operator be g_{max} and g_{min}, and their corresponding eigenstates be $|g_{max}\rangle$ and $|g_{min}\rangle$.
To maximise the variance of $\hat{G}$, we must choose our initial probe state such that it is a superposition of the $|g_{max}\rangle$ and $|g_{min}\rangle$ states. This is *exactly* what GHZ and N00N states do. Taking the example of a GHZ state:

$$|\psi_0\rangle = \frac{1}{\sqrt{2}}\Big(\,|g_{max}\rangle^{\otimes N} + |g_{min}\rangle^{\otimes N}\,\Big)$$

Now to find variance of $\hat{G}$

$$\langle\hat{G}\rangle = \langle\psi_0|\,\hat{G}\,|\psi_0\rangle = \frac{N}{2}(g_{max} + g_{min})$$

$$\langle\hat{G}^2\rangle = \langle\psi_0|\,\hat{G}^2\,|\psi_0\rangle = \frac{N^2}{2}(g_{max}^2 + g_{min}^2)$$

$$\mathrm{Var}(\hat{G}) = \langle\hat{G}^2\rangle - \langle\hat{G}\rangle^2 = \frac{N^2}{2}(g_{max}^2 + g_{min}^2) - \frac{N^2}{4}(g_{max} + g_{min})^2 = \frac{N^2}{4}(g_{max} - g_{min})^2$$

Thus, we obtain the QFI

$$I_Q = N^2(g_{max} - g_{min})^2$$

Therefore, from the QCRB

$$\mathrm{Var}(\theta) \geq \frac{1}{N^2(g_{max} - g_{min})^2} \propto \frac{1}{N^2}$$

Thus, the **Heisenberg Limit** has been achieved. For K independent measurements, we can say

$$\mathrm{Var}(\theta) \geq \frac{1}{KN^2}$$

To understand the difference, let's take a case where the HL isn't achieved. Consider the separable initial state

$$|\psi_0\rangle = \left(\frac{|g_{max}\rangle + |g_{min}\rangle}{\sqrt{2}}\right)^{\otimes N}$$

Since the individual states are now uncorrelated, the variance of $\hat{G}$ is simply the sum of the variances of all the $\hat{g}$ operators, i.e. $\mathrm{Var}(\hat{G}) = N{\cdot}\mathrm{Var}(\hat{g})$. Thus, we calculate

$$\langle\hat{g}\rangle = \langle\psi_0|\,\hat{g}\,|\psi_0\rangle = \frac{1}{2}(g_{max} + g_{min})$$

$$\langle \hat{g}^2 \rangle = \langle \psi_0 | \, \hat{g}^2 \, | \psi_0 \rangle = \frac{1}{2}(g_{max}^2 + g_{min}^2)$$

$$\text{Var}(\hat{g}) = \langle \hat{g}^2 \rangle - \langle \hat{g} \rangle^2 = \frac{1}{2}(g_{max}^2 + g_{min}^2) - \frac{1}{4}(g_{max} + g_{min})^2 = \frac{1}{4}(g_{max} - g_{min})^2$$

Thus, we obtain the QFI

$$I_Q = \text{Var}(\hat{G}) = N(g_{max} - g_{min})^2$$

Therefore, from the QCRB

$$\text{Var}(\theta) \geq \frac{1}{N(g_{max} - g_{min})^2} \propto \frac{1}{N}$$

Thus, the Heisenberg Limit has **not** been achieved.

2.3 References

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- For the KLD and Fisher Information relation, the reference is Dabak, Anand G. and Don H. Johnson. "Relations between Kullback-Leibler distance and Fisher information." (2002). http://dhj.rice.edu/files/2014/07/distance.pdf
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- The g_{max}, g_{min} stuff is from Giovannetti, Vittorio & Lloyd, Seth & Maccone, Lorenzo. (2006). Quantum Metrology. Physical review letters. 96. 010401. https://doi.org/10.1103/PhysRevLett.96.010401

3 Interferometric Quantum Sensing

3.1 Beam Splitters in Quantum Mechanics

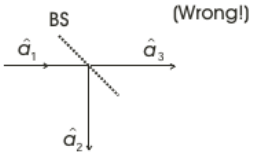


Fig. 6.4. Naïve, and incorrect, quantum mechanical depiction of a beam splitter.

- In the classical beam splitter, there's a single input beam that splits into two. This doesn't really work in the quantum case. The transformations don't preserve the required commutation relations, so we can't say that it's an accurate descriptor of a quantum beam splitter.
  The commutation relations that must be satisfied are:

$$\left[\hat{a}_i, \hat{a}_j^\dagger\right] = \delta_{ij} \qquad \left[\hat{a}_i, \hat{a}_j\right] = 0 = \left[\hat{a}_i^\dagger, \hat{a}_j^\dagger\right]$$

If we use the relations $\hat{a}_2 = r\hat{a}_1$ and $\hat{a}_3 = t\hat{a}_1$, r and t being the complex reflectance and transmittance of the beam splitter, then we can see that

$$\left[\hat{a}_2, \hat{a}_3^\dagger\right] = rt^* \left[\hat{a}_1, \hat{a}_1^\dagger\right] \neq 0$$

Hence, the required commutation relations aren't satisfied

- *Why use annihilation operators in place of amplitudes?*
If classical amplitude is $\mathcal{E}$ then intensity is $|\mathcal{E}|^2$. The corresponding observable in quantum mechanics is the number operator $\hat{n} = \hat{a}^\dagger\hat{a}$. So we can use annihilation operators as quantum analogues to classical wave amplitudes.

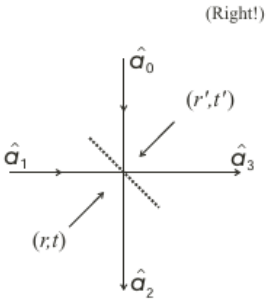


Fig. 6.5. The correct quantum-mechanical depiction of a beam splitter.

- The accurate descriptor has two inputs and two outputs. The transformations are:

$$\hat{a}_2 = r\hat{a}_1 + t'\hat{a}_0 \qquad \hat{a}_3 = t\hat{a}_1 + r'\hat{a}_0$$

These satisfy the commutation relations, as long as:

$$|r'| = |r|,\ |t| = |t'|,\ |r|^2 + |t|^2 = 1,\ r^*t' + r't'^* = 0, \text{ and } r^*t + r't'^* = 0$$

- So for a 50:50 beam splitter, the transformation relations will be:

$$\hat{a}_2 = \frac{1}{\sqrt{2}}(\hat{a}_1 + i\hat{a}_0) \qquad \hat{a}_3 = \frac{1}{\sqrt{2}}(i\hat{a}_1 + \hat{a}_0)$$

Remember that the reflected component gets phase shift $e^{i\pi/2} = i$

- Using these equations, we get the relation:

$$\hat{a}_1^\dagger = \frac{1}{\sqrt{2}}(i\hat{a}_2^\dagger + \hat{a}_3^\dagger)$$

- We will use the state $|n\rangle_i$ to denote n photons going into/coming out of the port corresponding to $\hat{a}_i$. We'll only be dealing with the single photon case, so n will either be 0 or 1. We also know that the action of the annihilation operators are as follows:

$$\hat{a}_i^\dagger |0\rangle_i = |1\rangle_i \qquad \hat{a}_i |1\rangle_i = |0\rangle_i$$

- We can clearly see that an zero input will give a zero output, i.e., $|0\rangle_0 |0\rangle_1 \xrightarrow{\text{BS}} |0\rangle_2 |0\rangle_3$
- Using the above equations, we can say:

$$|0\rangle_0 |1\rangle_1 = \hat{a}_1^\dagger |0\rangle_0 |0\rangle_1 = \frac{1}{\sqrt{2}}(i\hat{a}_2^\dagger + \hat{a}_3^\dagger) |0\rangle_0 |0\rangle_1 \xrightarrow{\text{BS}} \frac{1}{\sqrt{2}}(i\hat{a}_2^\dagger + \hat{a}_3^\dagger) |0\rangle_2 |0\rangle_3 = \frac{1}{\sqrt{2}}\big(i\,|1\rangle_2 |0\rangle_3 + |0\rangle_2 |1\rangle_3\big)$$

Similarly, we can say:

$$|1\rangle_0 |0\rangle_1 \xrightarrow{\text{BS}} \frac{1}{\sqrt{2}}\big(|1\rangle_2 |0\rangle_3 + i\,|0\rangle_2 |1\rangle_3\big)$$

This shows that when 1 photon enters through one of the input ports, the other port remaining empty, it has an equal probability of being transmitted or reflected.

3.2 Interferometry

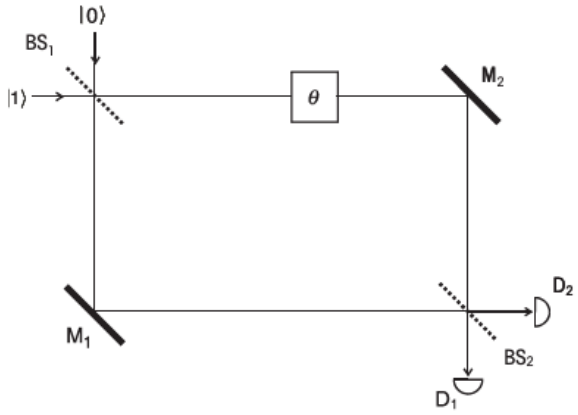


Fig. 6.7. A Mach–Zehnder interferometer with a single-photon input. The beam splitters BS_1 and BS_2 are 50:50, M_1 and M_2 are mirrors, and the box labeled θ represents the relative phase shift between the two arms.

- We'll use the convention that the states propagating along the counter-clockwise path precede those along the clockwise. So the input state shown in the above figure will be written as $|0\rangle\,|1\rangle$.
- The first beam splitter (BS_1) transforms the input as follows:

$$|0\rangle\,|1\rangle \xrightarrow{\text{BS}_1} \frac{1}{\sqrt{2}}\big(\,|0\rangle\,|1\rangle + i\,|1\rangle\,|0\rangle\,\big)$$

  The mirrors will contribute a factor of $e^{i\pi/2}$ to each term, which can be neglected as an overall phase.
- The phase shifter in the clockwise path provides a phase shift to the first term

$$\frac{1}{\sqrt{2}}\big(\,|0\rangle\,|1\rangle + i\,|1\rangle\,|0\rangle \xrightarrow{\theta} \frac{1}{\sqrt{2}}\big(e^{i\theta}\,|0\rangle\,|1\rangle + i\,|1\rangle\,|0\rangle\,\big)$$

  Only the first term gets the $e^{i\theta}$ factor, because only the photon going in the clockwise path will get phase-shifted.
- At the second beam splitter (BS_2), we have the following transformations:

$$|0\rangle\,|1\rangle \xrightarrow{\text{BS}_2} \frac{1}{\sqrt{2}}\big(\,|0\rangle\,|1\rangle + i\,|1\rangle\,|0\rangle\,\big) \qquad |1\rangle\,|0\rangle \xrightarrow{\text{BS}_2} \frac{1}{\sqrt{2}}\big(\,|1\rangle\,|0\rangle + i\,|0\rangle\,|1\rangle\,\big)$$

  Therefore, the final transformation is:

$$\frac{1}{\sqrt{2}}\big(e^{i\theta}\,|0\rangle\,|1\rangle + i\,|1\rangle\,|0\rangle\,\big) \xrightarrow{\text{BS}_2} \frac{1}{\sqrt{2}}\left(\frac{e^{i\theta}}{\sqrt{2}}\big(\,|0\rangle\,|1\rangle + i\,|1\rangle\,|0\rangle\,\big) + \frac{i}{\sqrt{2}}\big(\,|1\rangle\,|0\rangle + i\,|0\rangle\,|1\rangle\,\big)\right) = \frac{1}{2}\Big[(e^{i\theta}-1)\,|0\rangle\,|1\rangle + i(e^{i\theta}+1)\,|1\rangle\,|0\rangle\Big]$$

  Thus, we have the beam splitter output. We can now calculate probabilities.
- The probability that the state $|0\rangle\,|1\rangle$ is detected (i.e. that only D_2 fires) is given by

$$P_{01} = \left|\frac{1}{2}\big(e^{i\theta}-1\big)\right|^2 = \frac{1}{2}\big(1-\cos\theta\big)$$

- Similarly, the probability that the state $|1\rangle\,|0\rangle$ is detected (i.e. that only D_1 fires) is given by

$$P_{10} = \left|\frac{i}{2}\big(e^{i\theta}+1\big)\right|^2 = \frac{1}{2}\big(1+\cos\theta\big)$$

- Using these probability expressions, we can conduct trials and using the frequency of outcomes, estimate the value of θ upto some precision. Hence, they are useful in quantum interferometric sensing

3.3 Application

Now, MZI's are classical interferometers. For quantum parameter estimation, we generally prefer quantum interferometers such as GHZ-type state or N00N state interferometers. The reason for this is that they are entangled states, which enhances their resolution.
NOTE: Say we take a measurement utilising N subsystems that aren't entangled. For example, using MZI for N photons. In this case, the measurement precision $\Delta\theta$ scales according to the *Standard Quantum Limit* as follows:

$$\Delta\theta \geq \frac{1}{\sqrt{N}}$$

However, if the N subsystems are *entangled*, like in GHZ-type states, then the precision scales according to the *Heisenberg Limit* as follows:

$$\Delta\theta \geq \frac{1}{N}$$

So we see the systems with entanglement show better precision than non-entangled states.
END OF NOTE

- GHZ-type states are maximally entangled states. A general n-qubit GHZ state is given by

  $$|GHZ_N\rangle = \frac{1}{\sqrt{2}}\big(\,|0\rangle^{\otimes N} + |1\rangle^{\otimes N}\,\big)$$

  We see N-way entanglement here between N different distinguishable subsystems. Now when passed through the interferometer, instead of undergoing a θ phase shift, the state picks up a $N\theta$ phase shift (phase accumulation). Thus, the uncertainty scales as $\Delta\theta \geq \frac{1}{N}$ (Heisenberg Limit).

- The N00N state is a multi-photon entangled state. It is written as

  $$|\Psi\rangle = \frac{1}{\sqrt{2}}\big(\,|N\rangle\,|0\rangle + |0\rangle\,|N\rangle\,\big)$$

  So all N photons are in one of two optical modes.
  While GHZ states have 1 particle in each of the N subsystems, N00N states have N particles going into one of two subsystems.
  However, both of these states are still entangled. In GHZ states, the entanglement is between the N particles (particle entanglement), while in N00N states, the entanglement is between the two modes (mode entanglement).
  Similar to GHZ, entanglement in N00N states also means that the measurement uncertainly scales according to the Heisenberg Limit: $\Delta\theta \geq \frac{1}{N}$

Thus, we see how using entangled states in our interferometers greatly lowers our uncertainty. The problem with this is that states such as GHZ and N00N (with large N) are difficult to prepare. The larger the N, the better the precision, but the harder to prepare. This is one of the downsides to Quantum Interferometric Sensing.
Another downside is that we can only estimate parameters that can be implemented as phase-shift operator.

3.4 References

- Images and explanations from Section 3.1 and 3.2 are from: Gerry C, Knight P. *Introductory Quantum Optics.* Cambridge University Press; 2004. (Sections 6.2 and 6.3)